

Aayush

Noida, India

📞 +91 8851262529 • ✉ aayush002singh@gmail.com

Profile

Experienced AI-driven Software Engineer specializing in **Generative AI, Large Language Models (LLMs), and backend development**. Passionate about building **scalable AI systems, intelligent automation, and RAG pipelines** to optimize efficiency and enhance decision-making. Proficient in designing and deploying ML-driven solutions that drive innovation. Strong expertise in developing **high-performance backend architectures**, leveraging **asynchronous processing** and optimizing **RESTful APIs** for scalability, reliability, and seamless integration.

Experience

Shorthills AI

Software Engineer

Gurugram, India

Jun 2023 – Present

AI Studio <https://studio.shorthills.ai/>

- Spearheaded the development of an AI platform to accelerate AI/ML adoption and streamline automation.
- Built and optimized intelligent document processing tools like **DocLens** (extracting insights from financial/legal PDFs) and **ChatScribe** (AI-driven PDF chat assistant), increasing document processing efficiency by **35%**.
- Developed **QueryCraft**, an AI-powered tool that converts text inputs into SQL queries using **VannaAI**, achieving **85%** accuracy.
- Engineered a **scalable Django backend** with **Celery** for asynchronous task processing and optimizing document parsing.

Agent Assist

- Designed and implemented an AI-powered **Agent Assist system** to provide real-time support to customer service teams.
- Developed **Agentic RAG pipelines**, improving data accuracy and response quality by **30%**.
- Created a custom algorithm for fast and accurate data retrieval in Excel using **FuzzyWuzzy**, enhancing entity matching precision by **45%**.
- Reduced agent response times by **40%** and minimized resolution errors by **25%**.

Get Help Chatbot

- Developed an intent and entity recognition system to categorize user queries and extract key information.
- Generated **15,000 synthetic data samples** using dynamic prompts with GPT-4 to enhance LLM training.
- Achieved **97%** accuracy in recognition tasks by fine-tuning **Llama 3 8B Instruct** using **PEFT**.
- Optimized chatbot response efficiency, reducing manual query resolution time by **60%** and improving user experience.

AI Voice Chatbot

- Developed an AI-powered voice chatbot for automated customer interactions, improving response times by **50%**.
- Integrated speech-to-text and text-to-speech modules, achieving **92%** accuracy in voice recognition.

Publications

2023: Smart Charging: A Deployment Algorithm for Optimizing Wireless Rechargeable Sensor Networks

Proposed an optimized WRSN mobile charger scheduling algorithm, reducing energy loss by 12.3%.

<https://dl.acm.org/doi/abs/10.1145/3607947.3607959>

Education

Netaji Subhas University Of Technology (NSUT, Main Campus), Delhi

B.Tech in Mathematics and Computing

2019 – 2023

Technical Skills

AI/ML: Generative AI, LLM Fine-Tuning, RAG, LangChain, ML

Backend: Python, Django, FastAPI, REST APIs, Celery (Asynchronous Processing), Docker

Databases: Weaviate, FAISS, Qdrant, MySQL